Session 5 · Deck 5.A1

Perturbation: The Idea

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Build it on paper → solve it on the machine → push it to the frontier with AI

Where We Are

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Sep 24	Oct 1	Oct 8	Oct 15	Oct 22	Oct 29	Nov 5	Nov 12	Nov 19	Dec 3
Intro	HA · OLG	Num. DP	Numerics	Numerics	HA classic	ML	RL	Text	Agents
Growth	Policy	Python	I	II	The wall	Deep nets	HA + DL	LLMs	Projects
trunk	branches	the machine · classical methods			the wall	new tools		new data	colleagues

Build it on paper → solve it on the machine → push it to the frontier with AI

Block A — perturbation: 5.A1 the idea · 5.A2 the growth model to first order · 5.A3 which solution, higher order, distributions.

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Roadmap: Four Parts

This deck has no economics in it yet. It explains one idea — expand around a point you can solve — and the three pieces of calculus that make it work.

Part	Topic	The question it answers
1	What we are solving	What kind of object is "the solution of a model"?
2	The idea, on something you know	How can a series around a known point solve an unknown problem?
3	The three tools	What licenses the expansion, and where do the equations for its coefficients come from?
4	Inequalities	What if the model has a borrowing limit?

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By the end: "solve the model" will mean "compute a few derivatives at the steady state", and you will know exactly which theorem lets you say that.

1 · An Unknown Function Solves an Equation

Every model in this course asks for a function, not a number. Perturbation represents that function by a short list of derivatives at one point.

Write the equilibrium conditions as one equation,

$$H\Big(\underbrace{d}_{\text{unknown rule}}, \underbrace{x}_{\text{state}}, \underbrace{\theta}_{\text{parameters}}\Big) = 0,$$

where d is the **decision rule** — how much to consume, how much to save — as a function of the state. In Sessions 3–4 we stored d as a table of values on a grid and iterated.

Perturbation instead writes d as a **series around a point x_0 where the answer is known**:

$$d^n(x,\theta) = \sum_{i=0}^{n} a_i \,(x - x_0)^i,$$

and picks $a_0, \ldots, a_n$ so that $H = 0$ holds *up to order* n at x_0.

Idea

A Taylor series is a polynomial that matches a function and its slopes at one point. If we can find the slopes, we have the function nearby.

Example

The known point is the **deterministic steady state**, computed since 1.B1 in one line. In the growth model $d = (c, k')$ and $x = (k, z)$.

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1 · Six Steps, Once

The whole block is these six steps. Keep the table in view; each later deck does one or two of them.

	Step	Where
1	Write the equilibrium conditions as one system $F(\cdot\,;\chi) = 0$	5.A2
2	Choose the expansion point: the deterministic steady state, $z = 0$, $\chi = 0$	5.A2
3	Replace the policy functions by Taylor series around it	5.A2
4	Differentiate $F = 0$ there; each derivative is an equation for the coefficients	5.A2
5	Solve the linear systems — first order, then higher orders	5.A2, 5.A3
6	Check that the linear solution exists and is unique; ask how far it is valid	5.A3, 5.B4

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Idea

No grid, no fixed point, no maximization. Steps 4 and 5 are differentiation and linear algebra — which is the entire reason the method is fast.

Pitfall

Everything here is **local**: it describes the economy for small deviations from the expansion point and makes no claim beyond them.

χ **is the parameter that scales the shock, introduced in 5.A2.**

The Idea on a Number: $\sqrt{26}$ Without a Calculator

A hard problem is an easy problem plus a small change. Expand in the change.

You cannot take $\sqrt{26}$ in your head. You can take $\sqrt{25}$. So write

$$\sqrt{26} = \sqrt{25}\,\sqrt{1+\varepsilon}, \qquad \varepsilon = \tfrac{1}{25} = 0.04,$$

and expand only the part that carries the change:

$$\sqrt{1+\varepsilon} \approx \underbrace{1}_{\text{easy answer}} + \underbrace{\tfrac{\varepsilon}{2}}_{\text{1st-order fix}} - \underbrace{\tfrac{\varepsilon^2}{8}}_{\text{2nd-order fix}}.$$

First order gives 5.1, second order 5.099; the truth is 5.0990. **A problem you can solve, plus corrections in powers of the distance from it**: that is the whole method.

Example: error of $\sqrt{x}$ in %, further out

ε	x	1st order	2nd order
0.04	26	0.019	0.00038
0.5	37.5	2.1	0.49
1	50	6.1	2.8
3	100	25	31

Pitfall

Beyond $\varepsilon = 1$ an extra order makes it **worse**: the series converges only for $|\varepsilon| < 1$.

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Ledger `toy_perturbation.sqrt26`; $x = 25(1+\varepsilon)$.

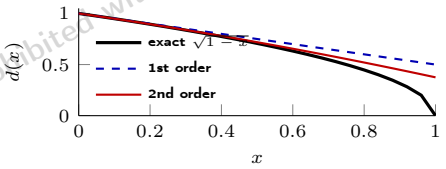
The Idea on a Function: Match Coefficients

Now the unknown is a function. Guess a series, substitute, and set every power's coefficient to zero: each step is a linear equation.

Find $d(x)$ with $H(d, x) = d(x)^2 + x - 1 = 0$. At $x = 0$ the answer is known: $d(0) = 1$. Guess $d(x) = a_0 + a_1 x + a_2 x^2 + \cdots$, substitute, collect powers of x:

$$\underbrace{(a_0^2 - 1)}_{x^0} + \underbrace{(2a_0a_1 + 1)}_{x^1} x + \underbrace{(a_1^2 + 2a_0a_2)}_{x^2} x^2 + \cdots \equiv 0.$$

This holds for *every* x, so **every coefficient is zero**: $a_0 = 1$, then $a_1 = -\frac{1}{2}$, then $a_2 = -\frac{1}{8}$. Each new unknown appears **linearly**, multiplied by $2a_0$, once the earlier ones are known. The truth is $d(x) = \sqrt{1 - x}$; these are its Taylor coefficients.



Idea

The multiplier of each new unknown is $2a_0 = \partial H / \partial d$ at the known point. If it were zero, nothing could be solved for: the hypothesis of the next theorem.

Ledger `toy_perturbation.matching`.

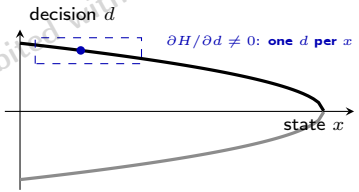
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Tool 1: The Implicit Function Theorem

Before expanding the decision rule in a series we need to know that a rule exists. One condition buys that, and more.

Idea

If the equation *responds* to the unknown at a point where you have a solution, then nearby there is exactly one solution per state, and it is smooth in the state.



The theorem. If $H(d_0, x_0, \theta_0) = 0$ and

$$\underbrace{\frac{\partial H}{\partial d}(d_0, x_0, \theta_0)}_{\text{response of the equation to the unknown}} \neq 0,$$

then near (x_0, θ_0) there is a unique $d = f(x, \theta)$ with $H(f, x, \theta) = 0$, as smooth as H: **existence**, **local uniqueness**, **differentiability**.

Example

In the dashed window the curve $H = d^2 + x - 1 = 0$ is a function of x: one d for each x, namely $d = \sqrt{1 - x}$.

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Two derivatives appear, and they point opposite ways: one must be non-zero, the other must be zero. They are derivatives of different functions.

One must be non-zero.

$$\frac{\partial H}{\partial d} \neq 0 \quad \text{at } (d_0, x_0, \theta_0).$$

This is solvability *in the d direction* — the hypothesis of the theorem. Here d is a free argument of H.

The other must be zero. Once $d = f(x, \theta)$ is the solution, substitute it in and call the result G:

$$G(x, \theta) \;:=\; H\big(f(x, \theta), x, \theta\big) \;\equiv\; 0.$$

G is zero for *every* (x, θ) in the window — it is the zero function. So its derivatives are zero too:

$$\frac{\partial G}{\partial x} = 0, \qquad \frac{\partial G}{\partial \theta} = 0.$$

Idea

H is a recipe with the decision as a free ingredient. G is what is left after the *right* decision has been put in: nothing. The derivative of nothing is nothing.

Example

$H = d^2 + x - 1$ has $\partial H/\partial d = 2d \neq 0$ at $(1, 0)$. With $f(x) = \sqrt{1 - x}$ put in, $G(x) = (1 - x) + x - 1 = 0$ for every x.

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Is the Hypothesis Met in the Growth Model? Yes

In the growth model H is the pair (Euler equation, resource constraint) and the unknown is $d = (c, k')$. Two economic facts make the condition hold.

With two equations and two unknowns, "$\partial H/\partial d \neq 0$" means the 2×2 matrix of responses is invertible:

$$\frac{\partial H}{\partial (c, k')} = \begin{pmatrix} \partial_c(\text{Euler}) & \partial_{k'}(\text{Euler}) \\ 1 & 1 \end{pmatrix}.$$

- $u''(c) < 0$: $\partial_c(\text{Euler}) = u''(c) \cdot (\ldots) \neq 0$ — concave utility makes the Euler equation respond to c at all.
- **The budget is linear** in (c, k'), coefficient 1 on each.

Together: full rank at the steady state. A singular matrix would mean one more unit of saving costs nothing.

Idea

This is the *static* condition: given tomorrow, today's (c, k') is pinned down. A dynamic model needs a second, separate check, that the linearized system has exactly one stable solution: Blanchard–Kahn, 5.A3. Neither implies the other.

Example

$u'' < 0$ has now done four jobs: a unique maximum (1.B2), one peak for golden section (4.A1), Santos's bound (3.A1), and local solvability.

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Tool 2 and 3: Taylor's Theorem, and the Chain Rule

Taylor turns the unknown function into unknown numbers; the chain rule on $G \equiv 0$ pins each one down.

Tool 2, Taylor. Near the reference point, $f = f_0 + f_x(x - x_0) + f_\theta(\theta - \theta_0) + \cdots$, so "find f" becomes "find the numbers $f_x, f_\theta, f_{xx}, \ldots$"; the theorem also bounds what a truncation leaves out.

Tool 3, the chain rule on $G \equiv 0$. Differentiate $G(x, \theta) = H\big(f(x, \theta), x, \theta\big)$ in x:

$$\underbrace{\frac{\partial H}{\partial d}}_{\text{known}} \underbrace{f_x}_{\text{unknown}} + \underbrace{\frac{\partial H}{\partial x}}_{\text{known}} = 0 \implies f_x = -\frac{\partial H/\partial x}{\partial H/\partial d},$$

and the same with θ. "Known" means evaluated at the reference point; the IFT hypothesis $\partial H/\partial d \neq 0$ makes the division legal (with $d = (c, k')$ it is the 2×2 matrix of the last frame, and "divide" means "solve the system").

Then repeat. One more derivative: f_{xx} appears linearly, times the same $\partial H/\partial d$, with f_x known. Each order is a linear solve; only the first has a choice (which root; 5.A3).

Idea

The $\sqrt{1-x}$ frame again, with derivatives in place of powers of x: match coefficients order by order, and the multiplier of every new unknown is $\partial H/\partial d$.

Example

$H = d^2 + x - 1$ at $(d, x) = (1, 0)$: $\partial H/\partial d = 2$, $\partial H/\partial x = 1$, so $f_x = -\frac{1}{2}$ — the a_1 found by matching powers.

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The Toolkit, Collected

Five tools. Three are calculus you have seen, one you used in Sessions 2 and 4, and one is new to this session.

1. **Implicit function theorem** — licenses writing d as a function of (x, θ) near the reference point.
2. **Taylor's theorem** — expands that function, turning "solve for d" into "solve for its derivatives".
3. **Differentiating an identity** — if $G \equiv 0$ then *every* derivative of G is zero. This is where the equations come from.
4. **An eigenvalue condition** — Blanchard–Kahn, which decides whether the linear system we end up with has one stable solution, many, or none. *New, in 5.A3.*
5. **Complementarity reformulation** — Fischer–Burmeister, when the model has inequalities. *Already used in 2.A1 and 4.A2.*

Only item 4 is new. Items 1–3 are one page of calculus; item 5 you have met twice.

Idea

Perturbation is **local**. It answers "how does the economy behave for small deviations from the expansion point?" and nothing else. Block B is for everything else.

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When the Model Has Inequalities

The method needs a system of equalities. A borrowing limit gives an inequality and a complementarity condition instead; one function turns the three into one equation.

Kuhn–Tucker conditions have the shape

$$a \geq 0, \qquad h \geq 0, \qquad a\,h = 0,$$

"both non-negative, at least one zero" (a a multiplier, h the slack of the constraint). The **Fischer–Burmeister** function

$$\Psi^{FB}(a,h) \;=\; a + h - \sqrt{a^2 + h^2} \;=\; 0$$

is zero exactly on that set. Check the cases:

- $a > 0$, $h = 0$: $a + 0 - a = 0$. $\quad a = 0$, $h > 0$: $0 + h - h = 0$. $\quad (0,0)$: 0.
- $a > 0$ and $h > 0$: $\sqrt{a^2 + h^2} < a + h$, so $\Psi^{FB} > 0$ — not zero.

Idea

After the rewriting the equilibrium system is again $F(\cdot) = 0$, so the same workflow — expand, differentiate, match — applies without change.

Example

2.A1's borrowing constraint $a' \geq -\underline{a}$ with its multiplier; 4.A2 solved the same equation by Newton. Same notation here.

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Takeaways

- **Part 1.** The solution of a model is a function. Perturbation represents it by a few **derivatives at the steady state** instead of a table of values on a grid.
- **Part 2.** $\sqrt{26} = \sqrt{25}\sqrt{1+0.04}$: an easy problem plus corrections in powers of the distance. For a function, guess a series and **match coefficients**; each new coefficient solves a linear equation.
- **Part 3.** The implicit function theorem licenses the series ($\partial H/\partial d \neq 0$). Substituting the solution makes $G \equiv 0$, so every derivative of G is zero — and by the chain rule each one is a linear equation for an unknown slope. The procedure is recursive, order by order.
- **Part 4.** Inequalities become equalities through the Fischer–Burmeister function, and the workflow is unchanged.

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Bridge: A Model, and One Extra Parameter

So far this has been calculus with no economics in it. The next deck puts a model underneath: the stochastic growth model with log utility and full depreciation — the one model whose policy we know exactly, so every step can be checked against the answer.

It also introduces the trick that makes uncertainty tractable: an artificial parameter that **scales the shock**, so that "no risk" is a point we can expand around.

Next (5.A2)

The model and its closed form; the parameter χ; the expansion written out; the equations for its coefficients, and why two of the six first-order unknowns are exactly zero.

Homework 1 is due next Thursday, Oct 29. **Homework 2** goes out the same day.

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